

Adventist University of Central Africa

Master's in Big Data Analytics

Chapter 2 — Statistical Learning

Exercises 1 and 10

Introduction to Statistical Learning with Applications in R

Programme: MSc Big Data Analytics
Course Title: Data Mining
Course Code: MSDA 9124
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September 2026

1 Exercise 1 — Flexible vs. Inflexible Methods

For every part below, the comparison comes down to one identity: the expected test mean squared error (MSE) at a new point x_0 decomposes as

$$E[(y_0 - \hat{f}(x_0))^2] = \text{Var}(\hat{f}(x_0)) + [\text{Bias}(\hat{f}(x_0))]^2 + \text{Var}(\epsilon)$$

A **flexible** method (e.g. splines, high-degree polynomials, kNN with small k) can bend to follow intricate patterns in the training data, so it tends to have **low bias** but **high variance** — it changes a lot if you swap in a different training sample. An **inflexible** method (e.g. linear regression) is the opposite: it can't bend much, so it has **higher bias** but **low variance** — it barely changes from one training sample to the next. Every part of this question is really asking: *in this scenario, does reducing bias (going flexible) buy you more than the variance it costs?*

1.1 (a) n extremely large, p small

Better: flexible.

Think of it like sketching a coastline. With only a handful of GPS points (small n), you can only draw a straight or gently curved line — try to trace every inlet and you'll just be connecting noise. But hand a cartographer thousands of closely-spaced points (large n), and tracing every inlet and headland becomes not just possible but *more* accurate than forcing a straight line through them.

Formally: a large n relative to p means there is enough data to estimate a complex $\hat{f}$ **without** the fit becoming unstable — the $\text{Var}(\hat{f}(x_0))$ term stays under control even as flexibility goes up, because each local region of the predictor space is still densely populated with observations. Meanwhile an inflexible method leaves the bias term unnecessarily large when there's no need for it — there's enough data to support a far better approximation of the true f . So the bias reduction from flexibility comes almost for free.

1.2 (b) p extremely large, n small

Better: inflexible.

This is the mirror image of (a), and it's the classic *curse of dimensionality*: imagine trying to furnish a house (small n = few pieces of furniture) that keeps gaining extra rooms (large p = more predictors). Spread thin across so many rooms, the house starts to look empty no matter how you arrange things — there just isn't enough "stuff" to fill the space convincingly.

Formally: with few observations spread across a high-dimensional predictor space, each region of that space is sparsely populated. A flexible method will latch onto whatever idiosyncratic pattern happens to exist among those few points — $\text{Var}(\hat{f}(x_0))$ explodes — and that variance inflation outweighs any bias it saves. An inflexible method, constrained to a simple global shape, can't overfit the sparse data in the same way, so it generalizes better despite its higher bias.

1.3 (c) Relationship between predictors and response is highly non-linear

Better: flexible.

Picture trying to trace a winding mountain road using only straight segments of highway — you’ll cut every corner, and the further the road curves, the worse your straight-line approximation gets. A flexible method is the one willing to bend with the road.

Formally: if the true f is highly non-linear, an inflexible method like linear regression has a *structural* limitation — no matter how much data you give it, it cannot represent the curvature, so $[\text{Bias}(\hat{f}(x_0))]^2$ stays large regardless of n . A flexible method can actually approximate the curvature, driving bias down. As long as n is reasonably sized, the resulting increase in variance is smaller than the bias it eliminates, so total expected test MSE drops.

1.4 (d) Variance of the error terms $\sigma^2 = \text{Var}(\epsilon)$ is extremely high

Better: inflexible.

Think of trying to hear a friend’s voice (the true signal f) over a room full of loud static (the irreducible noise ϵ). Leaning in and straining to catch every fluctuation in the sound (a flexible method) mostly means you start “hearing” patterns in the static itself, not the voice. You’re better off just catching the general gist.

Formally: $\text{Var}(\epsilon)$ is **irreducible** — it sits in the MSE decomposition as a constant floor that *no* method, however flexible, can lower. A flexible method doesn’t just fail to reduce it — it actively makes things worse, because it will interpret large random fluctuations in the training data as if they were real signal and fit to them, inflating $\text{Var}(\hat{f}(x_0))$ on top of the noise floor that was already there. An inflexible method resists chasing that noise, so its total expected test MSE stays closer to the irreducible floor.

2 Exercise 10 — The Boston Housing Data Set

```
library(ISLR2)
data(Boston)
```

2.1 (a) Rows, columns, and what they represent

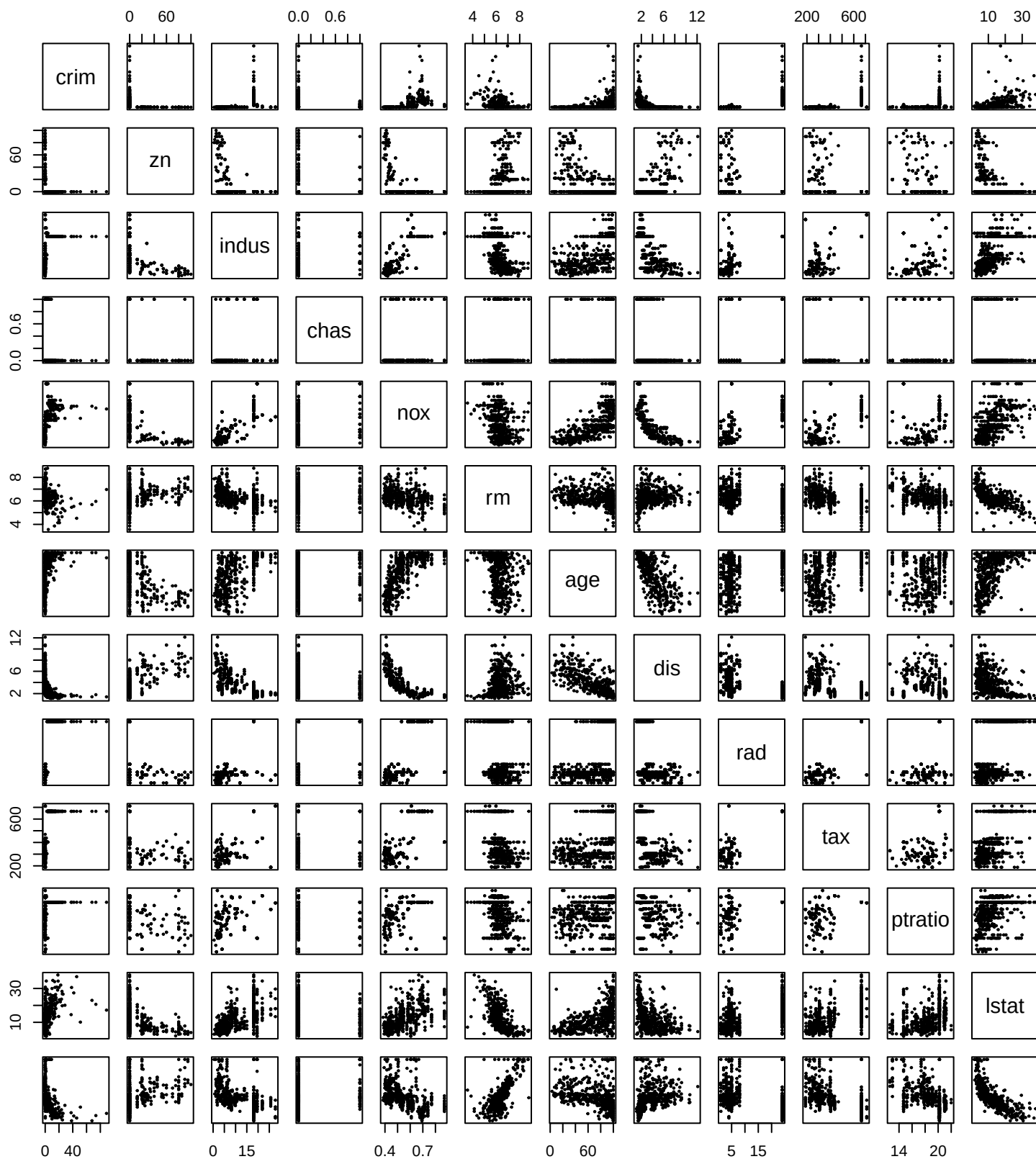
```
dim(Boston)
```

```
## [1] 506 13
```

The data set has **506 rows and 13 columns**. Each **row** is a single census tract (a small geographic subdivision) in the Boston area. Each **column** is a variable recorded for that tract — for example **crim** (per capita crime rate), **rm** (average rooms per dwelling), **ptratio** (pupil-teacher ratio), and **medv** (median value of owner-occupied homes in \$1000s), among others. So this is not 506 individual houses — it's 506 neighborhoods, each summarized by 13 socioeconomic and structural statistics.

2.2 (b) Pairwise scatterplots

```
pairs(Boston, pch = 20, cex = 0.4)
```



A full 13×13 grid is dense, so it helps to know what to look for rather than eyeball all 78 panels equally. A few patterns stand out visually and are confirmed by the correlations pulled out in part (c) below:

- **lstat vs. medv**: a clear negative, curved (not perfectly linear) relationship — tracts with a higher share of lower-status population have systematically lower home values.
- **rm vs. medv**: a positive relationship — more rooms per dwelling tends to mean higher home value, which makes intuitive sense (bigger houses cost more).
- **nox vs. indus** and **nox vs. age**: positive relationships — tracts with more industrial land use, and with older housing stock, tend to have higher nitrogen oxide pollution.
- **tax vs. rad**: a striking pattern where many tracts cluster at one very high value of **rad** (the accessibility-to-highways index) paired with a high, fairly constant **tax** rate — visible as a dense vertical/ horizontal band rather than a smooth trend.
- **dis vs. nox, age, indus**: negative relationships — tracts farther from Boston's employment centers tend to be less industrial, have newer housing, and lower pollution (consistent with them being more suburban).

2.3 (c) Predictors associated with per-capita crime rate

```
sort(cor(Boston)[, "crim"])
```

```
##          medv          dis          rm          zn          chas          ptratio
## -0.38830461 -0.37967009 -0.21924670 -0.20046922 -0.05589158  0.28994558
##          age          indus          nox          lstat          tax          rad
##  0.35273425  0.40658341  0.42097171  0.45562148  0.58276431  0.62550515
##          crim
##  1.00000000
```

Yes — several predictors show a meaningful relationship with **crim**:

- **rad** ($r \approx 0.63$) and **tax** ($r \approx 0.58$) are the strongest — tracts with better highway accessibility and higher property tax rates tend to have notably higher crime rates. This is really one pattern more than two, since **rad** and **tax** are themselves highly correlated (visible as that dense cluster in part (b)) — a small subset of tracts with maxed-out **rad** and **tax** values are dragging crime up with them.
- **lstat** ($r \approx 0.46$) and **nox** ($r \approx 0.42$) are moderately positively associated — tracts with more lower-status population and more air pollution tend to have higher crime.
- **medv** ($r \approx -0.39$) and **dis** ($r \approx -0.38$) are moderately negatively associated — higher home values and greater distance from employment centers both correspond to lower crime.

Note these are all linear (Pearson) correlations; the true relationships may well be non-linear, so these numbers indicate *association*, not necessarily the full shape of the relationship.

2.4 (d) Notably high crime, tax, or pupil-teacher tracts; ranges

```
sapply(Boston[, c("crim", "tax", "ptratio")], range)
```

```
##          crim tax ptratio
## [1,]  0.00632 187    12.6
## [2,] 88.97620 711    22.0
```

```
sapply(Boston[, c("crim", "tax", "ptratio")], summary)
```

```
##          crim          tax          ptratio
```

```
## Min.      0.006320 187.0000 12.60000
## 1st Qu.   0.082045 279.0000 17.40000
## Median    0.256510 330.0000 19.05000
## Mean      3.613524 408.2372 18.45553
## 3rd Qu.   3.677083 666.0000 20.20000
## Max.      88.976200 711.0000 22.00000
```

- **crim** ranges from **0.006 to 88.98** — an enormous spread. The median tract has a crime rate near 0.26, so a handful of tracts (crime rate in the 40s–80s) are extreme outliers pulling the mean well above the median — most tracts are low-crime, with a small number of very high-crime tracts.
- **tax** ranges from **187 to 711** (per \$10,000 of property value). There’s a visible cluster of tracts sitting right at the maximum (666), the same cluster seen in the **tax-rad** pattern from part (b) — so while the *range* is wide, a meaningful subgroup of tracts is bunched at the top of it.
- **ptratio** ranges from **12.6 to 22.0** — a comparatively narrow spread on an absolute scale, though a pupil-teacher ratio of 22 vs. 12.6 is still a substantively large difference in school resourcing per student.

2.5 (e) Tracts bounding the Charles River

```
sum(Boston$chas)
```

```
## [1] 35
```

35 of the 506 census tracts bound the Charles River (`chas == 1`).

2.6 (f) Median pupil-teacher ratio

```
median(Boston$ptratio)
```

```
## [1] 19.05
```

The median pupil-teacher ratio across tracts is **19.05**.

2.7 (g) Tract with the lowest median home value

```
which.min(Boston$medv)
```

```
## [1] 399
```

```
Boston[which.min(Boston$medv), ]
```

```
##      crim  zn  indus chas   nox   rm age   dis rad tax ptratio lstat medv
## 399 38.3518  0  18.1    0 0.693 5.453 100 1.4896  24  666    20.2 30.59    5
```

Tract **399** has the lowest `medv` at **\$5,000** — far below the data set’s minimum-to-typical range (the next visualization makes the point concretely: this tract sits at the very bottom of the `medv` distribution while most other predictors sit near the *top* of their own ranges).

```
idx <- which.min(Boston$medv)
sapply(names(Boston), function(v) round(100 * mean(Boston[[v]] <= Boston[idx, v]), 1))
```

```
##      crim      zn   indus   chas    nox    rm    age    dis    rad    tax
##    98.8    73.5   88.7   93.1   85.8    7.7   100.0    5.7   100.0   99.0
## ptratio   lstat   medv
##    88.9    97.8    0.4
```

Interpreting these as percentile ranks within the full data set, tract 399 is a striking outlier profile: it sits at the **100th percentile for age** (essentially all owner-occupied units built before 1940) and **rad** (maximum highway accessibility), the **99th percentile for tax**, the **98.8th percentile for crim**, and the **97.8th percentile for lstat** — while its **rm** (rooms per dwelling) is only at the 7.7th percentile and its **dis** (distance to employment centers) at the 5.7th percentile. In short: old housing stock, very high crime and tax burden, small homes, low-status population, and close to the urban core — a textbook combination for depressed home values, and consistent with the associations found in part (c).

2.8 (h) Tracts averaging more than seven / eight rooms

```
sum(Boston$rm > 7)
```

```
## [1] 64
```

```
sum(Boston$rm > 8)
```

```
## [1] 13
```

```
summary(Boston[Boston$rm > 8, ])
```

```
##      crim      zn   indus   chas
## Min.   :0.02009   Min.   : 0.00   Min.   : 2.680   Min.   :0.0000
## 1st Qu.:0.33147   1st Qu.: 0.00   1st Qu.: 3.970   1st Qu.:0.0000
## Median :0.52014   Median : 0.00   Median : 6.200   Median :0.0000
## Mean   :0.71880   Mean   :13.62   Mean   : 7.078   Mean   :0.1538
## 3rd Qu.:0.57834   3rd Qu.:20.00   3rd Qu.: 6.200   3rd Qu.:0.0000
## Max.   :3.47428   Max.   :95.00   Max.   :19.580   Max.   :1.0000
##      nox      rm      age      dis
## Min.   :0.4161   Min.   :8.034   Min.   : 8.40   Min.   :1.801
## 1st Qu.:0.5040   1st Qu.:8.247   1st Qu.:70.40   1st Qu.:2.288
## Median :0.5070   Median :8.297   Median :78.30   Median :2.894
## Mean   :0.5392   Mean   :8.349   Mean   :71.54   Mean   :3.430
## 3rd Qu.:0.6050   3rd Qu.:8.398   3rd Qu.:86.50   3rd Qu.:3.652
## Max.   :0.7180   Max.   :8.780   Max.   :93.90   Max.   :8.907
##      rad      tax      ptratio      lstat      medv
## Min.   : 2.000   Min.   :224.0   Min.   :13.00   Min.   :2.47   Min.   :21.9
## 1st Qu.: 5.000   1st Qu.:264.0   1st Qu.:14.70   1st Qu.:3.32   1st Qu.:41.7
## Median : 7.000   Median :307.0   Median :17.40   Median :4.14   Median :48.3
## Mean   : 7.462   Mean   :325.1   Mean   :16.36   Mean   :4.31   Mean   :44.2
## 3rd Qu.: 8.000   3rd Qu.:307.0   3rd Qu.:17.40   3rd Qu.:5.12   3rd Qu.:50.0
## Max.   :24.000   Max.   :666.0   Max.   :20.20   Max.   :7.44   Max.   :50.0
```

64 tracts average more than seven rooms per dwelling, and **13** average more than eight.

The 13 tracts with **rm > 8** are a notably *advantaged* subset compared to the data set as a whole:

their median `crim` (≈ 0.52) and `lstat` (≈ 4.14) are both far below the full data set's medians, and their median `medv` (\$48,300) sits near the top of the overall home-value range (recall the overall `medv` maximum is \$50,000, likely censored at that value). In other words, large-home tracts tend to combine low crime, low lower-status population share, and high home values — consistent with the negative `lstat-medv` and positive `rm-medv` relationships seen in part (b).